Software Test Plan (STP) - Smart Complaint Management System

Project: Smart Complaint Management System
Team: Team 8
Version: 1.0
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1. Introduction

Purpose: This document defines the test plan for the Smart Complaint Management System v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing user registration and authentication, complaint submission and tracking, evidence upload, department assignment, complaint resolution and closure, notifications, reporting, and citizen feedback.

Scope: Testing covers citizen registration and authentication, complaint submission with evidence, complaint tracking, status history, department assignment and reassignment, complaint resolution and closure, status notifications, admin dashboard/reporting, and citizen feedback. External email/SMS gateway internal infrastructure is excluded.

References: Smart Complaint Management System SRS v1.0, Design Specifications v1.0, Requirements Traceability Matrix (RTM).

Definitions: SCMS (Smart Complaint Management System), UI (User Interface), API (Application Programming Interface), DB (Database), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), RBAC (Role-Based Access Control), SLA (Service-Level Agreement).

2. Test Items

- User Registration & Authentication module
- Complaint Submission & Evidence Upload module
- Complaint Tracking & Status History module

- Department Assignment & Reassignment module
- Complaint Resolution & Closure module
- Notification Service
- Admin Dashboard & Reporting module
- Citizen Feedback module

3. Features to be Tested

Features mapped to SRS requirement IDs:

- SCMS-F-001: Citizen account registration with name, email/mobile number, and password
    - SCMS-F-002: User authentication with valid credentials
    - SCMS-F-003: Role-based access for Citizen, Department Staff, and Admin
    - SCMS-F-004: Complaint submission with category, description, and location
    - SCMS-F-005: Upload supporting image or PDF evidence
    - SCMS-F-006: Automatic generation of unique complaint tracking ID
    - SCMS-F-007: Complaint tracking using tracking ID
    - SCMS-F-008: Chronological complaint status history and remarks
    - SCMS-F-009: Complaint search and filtering by ID, category, status, and date
    - SCMS-F-010: Automatic complaint assignment to the responsible department
    - SCMS-F-011: Complaint reassignment by Admin
    - SCMS-F-012: Complaint status updates and resolution remarks
    - SCMS-F-013: Complaint closure after entering resolution details
    - SCMS-F-014: Citizen notification when complaint status changes
    - SCMS-F-015: Admin reports and citizen feedback after complaint closure
    - SCMS-NF-001: Complaint submission and status response time ≤ 3 seconds for 95% of requests
    - SCMS-NF-002: 99.5% monthly availability
    - SCMS-NF-003: Support 10,000 registered citizens and 1,000 new complaints per day
    - SCMS-NF-004: WCAG 2.1 Level AA accessibility where applicable
    - SCMS-NF-005: Complaint records and audit logs retained for at least five years
    - SCMS-SR-001: TLS 1.2+ for client-server communication
    - SCMS-SR-002: Secure salted adaptive password hashing
    - SCMS-SR-003: Server-side role-based access control
    - SCMS-SR-004: Secure session expiry and cookie configuration

- SCMS-SR-005: Audit logging of Admin and Staff actions

4. Features Not to be Tested

- Internal infrastructure of third-party email/SMS gateways
- Physical hardware or infrastructure outside the SCMS application
- Browser internal implementation
- External services not directly controlled by the SCMS application
- Database server hardware and operating-system internals

5. Test Approach / Strategy

Levels:

- Unit tests (module-level: authentication, complaints, tracking, routing, notifications, reporting)
- Integration tests (web application, backend API, database, file storage, and notification service)
- System tests (end-to-end SCMS workflows)
- Acceptance tests (UAT with citizens, department staff, and administrators)

Types:

- Functional testing (registration, complaints, tracking, assignment, resolution, notifications, reports, feedback)
- Regression testing (after each module integration and major change)
- Performance testing (response time, concurrent users, complaint volume)
- Usability testing (citizen workflow clarity and ease of complaint submission/tracking)
- Accessibility testing (WCAG 2.1 Level AA requirements)

Entry Criteria: Stable build delivered, test data (dummy users, complaints, departments, and status records) available, and test environment ready.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all acceptance criteria satisfied, and all high-priority requirements verified.

5.1 Security Validation

- Validate password hashing and confirm that no plaintext passwords are stored.
    - Verify TLS 1.2 or higher for all client-server communication.
    - Validate role-based access control for Citizen, Department Staff, and Admin.

- Perform input validation and fuzzing for registration, complaint details, tracking IDs, and file uploads.
- Validate session expiry after 15 minutes of inactivity and Secure, HttpOnly, and SameSite cookie settings.
- Verify audit logging for Admin and Staff actions.
- Perform security testing of authentication, complaint access, administrative functions, and protected APIs.

6. Test Environment

Hardware: Standard server for hosting the SCMS web application and database; client devices including desktop, laptop, and mobile devices for browser testing.

Software: SCMS web application v1.0, Python Flask/Django or PHP Laravel backend, MySQL/PostgreSQL database, modern browsers including Chrome, Firefox, and Edge.

Tools: Selenium (UI automation), Postman (API testing), JMeter (performance/load testing), pytest/Django test framework (unit testing), Jira/GitHub Issues (defect tracking).

Test Data: Dummy citizen, staff, and admin accounts; sample complaints across different categories and locations; complaint status histories; sample image/PDF attachments; department assignments; resolution records; and feedback records.

7. Test Schedule

- Milestones:
    - Test case design: 17-09-2026
    - Environment setup: 18-09-2026
    - Test execution start: 19-09-2026
    - Test execution end: 23-09-2026
    - UAT: 24-09-2026

8. Test Deliverables

- Test Plan (this document)
    - Test Cases (manual & automated)
    - Test Scripts
    - Test Data
    - Test Execution Logs

- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Rakshita Basappagoud Biradar	Prepare plan, coordinate execution
Test Engineer	Reet Anand	Design & execute test cases, log defects
Developer	Rehaan Jose Mathew	Support defect fixes and triage
Product Owner	Rayudu Meghana	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from development team
Test environment downtime	Maintain a backup test environment
Incomplete or inaccurate dummy complaint data	Prepare and validate test data before execution
Notification service dependency on email/SMS gateway	Use mock/stub gateway for testing without live sends
Large complaint volume affecting performance	Prepare representative load-test data and execute performance testing
Incorrect role permissions	Perform dedicated RBAC and security testing for all user roles

11. Assumptions & Dependencies

- The SCMS backend and web application will be stable and available during test cycles.
- Test data including citizen accounts, complaints, departments, and status records will be provided or generated before execution.
- The database instance will be accessible throughout the test period.
- File storage will be available for testing complaint attachments.
- Email/SMS notification services will be available or replaced with mock services during testing.
- Modern desktop and mobile browsers will be available for compatibility testing.

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment unavailable for more than 4 hours
- Build is too unstable and blocks more than 30% of test cases
- Critical application functionality required for testing is unavailable

Resume testing if:

- Blocking defects are resolved
- Environment stabilized
- A stable build is deployed and smoke testing is successful

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases. Example:

- SCMS-F-001 (Register account) → TC-Auth-01
- SCMS-F-004 (Lodge complaint) → TC-Lodge-01
- SCMS-F-005 (Attach evidence) → TC-Lodge-02
- SCMS-F-007 (Track complaint) → TC-Track-01
- SCMS-F-010 (Auto assignment) → TC-Assign-01
- SCMS-F-013 (Close complaint) → TC-Admin-02
- SCMS-F-014 (Status notification) → TC-Notify-01
- SCMS-NF-001 (Response time) → TC-Perf-01
- SCMS-SR-003 (Role-based access control) → TC-Sec-03
- SCMS-SR-005 (Audit log) → TC-Sec-05

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage
- % high-priority requirements verified

Reports:

- Daily execution status

- Defect status report
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Rakshita Basappagoud Biradar	Rakshita Basappagoud Biradar – 12/09/2026
Dev Lead	Rehaan Jose Mathew	Rehaan Jose Mathew – 12/09/2026
Product Owner	Rayudu Meghana	Rayudu Meghana – 12/09/2026